

Lecture 1

Probability Essentials

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(Edited from the lecture notes by Brian J. Reich, NC State)



Review of probability

- ▶ In Frequentist or classical statistics, we assume that the data $\mathbf{Y}$ follows a model that involves some unknown fixed parameter θ .
- ▶ In Bayesian statistics, we also assume a probability distribution (prior) for θ .
- ▶ The most important thing in Bayesian statistics is computing the posterior, i.e., the distribution of the parameters θ after observing the data $\mathbf{Y}$.
- ▶ This is the conditional distribution of θ given $\mathbf{Y}$.
- ▶ Therefore, we need to review the probability concepts that lead to the conditional distribution of one variable conditioned on another.
 1. Probability mass (PMF) and density (PDF) functions (Lecture 1)
 2. Joint distributions (Lecture 2)
 3. Marginal and conditional distributions (Lecture 2)

Random variables

- ▶ A random variable is generally denoted by X (capital script).
- ▶ We want to compute the probability of X taking a specific value x (lowercase script).
- ▶ This is denoted by $\text{Prob}(X = x)$.
- ▶ We also might want to compute the probability of X being in a set $\mathcal{A}$.
- ▶ This is denoted by $\text{Prob}(X \in \mathcal{A})$.
- ▶ The set of possible value that X can take on is called its support ($\mathcal{S}$).

Random variables - example

Example 1: X is the outcome of rolling a die.

- ▶ The support is $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$. Let $\mathcal{A} = \{2, 3\}$.
- ▶ $\text{Prob}(X = 5) = 1/6$, and $\text{Prob}(X \in \mathcal{A}) = 2/6 = 1/3$.

Example 2: X is the weight of a newborn baby.

- ▶ The support is $\mathcal{S} = (0, \infty)$.
- ▶ $\text{Prob}(X \in (0, \infty)) = 1$.

Example 3: X is the amount of rainfall measured at a weather station.

- ▶ The support is $\mathcal{S} = [0, \infty)$.
- ▶ $\text{Prob}(X \in [0, \infty)) = 1$ and $\text{Prob}(X = 0) \in [0, 1]$.

What is probability?



Academician A. N. Kolmogorov, S. N. Bose, P. C. Mahalanobis at a Special Convocation held to confer Honorary degree of Doctor of Science of the I.S.I. to Kolmogorov on 28 April, 1962

What is probability?

Objective (associated with a Frequentist)

- ▶ $\text{Prob}(X = x)$ as a purely mathematical statement.
- ▶ If we repeatedly sampled X , the the proportion of draws equal to x converges to $\text{Prob}(X = x)$.

Subjective (associated with Bayesian)

- ▶ $\text{Prob}(X = x)$ represents an individual's degree of belief.
- ▶ Often quantified as the amount an individual would be willing to wager that X will be x .

A Bayesian analysis makes use of both of these concepts.

What is uncertainty?



Figure: Copyright: ISI Kolkata archive

What is uncertainty?

Aleatoric uncertainty (likelihood)

- ▶ Uncontrollable randomness in the experiment.
- ▶ For example, the results of a fair coin flip can never be predicted with certainty.

Epistemic uncertainty (prior/posterior)

- ▶ Uncertainty about a quantity that could theoretically be known.
- ▶ For example, if we flipped a coin infinitely-many times we could know the true probability of a head.

A Bayesian analysis makes use of both of these concepts.

Probability versus statistics

Probability is the forward problem

- ▶ We assume we know how the data are being generated and compute the probability of events.
- ▶ For example, what is the probability of flipping 5 straight heads if the coin is fair?

Statistics is the inverse problem

- ▶ We use data to learn about the data-generating mechanism.
- ▶ For example, if we flipped five straight head, can we conclude the coin is biased?

Any statistical analysis obviously relies on probability.

Univariate distributions

- ▶ We often distinguish between **discrete** and **continuous** random variables (let us forget the discontinuous case for the sake of simplicity).
- ▶ The random variable X is discrete if its support $\mathcal{S}$ is countable
- ▶ Examples:
 - $X \in \{0, 1, 2, 3\}$ is the number of successes in 3 trials.
 - $X \in \{0, 1, 2, \dots\}$ is the number users that visit a website.

Univariate distributions

- ▶ We often distinguish between **discrete** and **continuous** random variables.
- ▶ The random variable X is continuous if its support $\mathcal{S}$ is uncountable.
- ▶ Examples with $\mathcal{S} = (0, \infty)$:
 - $X > 0$ is weight of a baby.
 - $X > 0$ is the wind speed.
- ▶ Examples with $\mathcal{S} = \mathbb{R}$:
 - X is the recorded temperature at a weather station.
 - X is the surge in the share price of a company.

Discrete univariate distributions

- ▶ If X is discrete we describe its distribution with its **probability mass function** (PMF).
- ▶ The PMF is $f(x) = \text{Prob}(X = x)$.
- ▶ The domain ($\mathcal{S}$) of X is the set of x with $f(x) > 0$.
- ▶ We must have $f(x) \geq 0$ and $\sum_{x \in \mathcal{S}} f(x) = 1$.
- ▶ The mean is $E(X) = \sum_{x \in \mathcal{S}} xf(x)$.
- ▶ The variance is $V(X) = \sum_{x \in \mathcal{S}} [x - E(X)]^2 f(x)$.

Parametric families of distributions

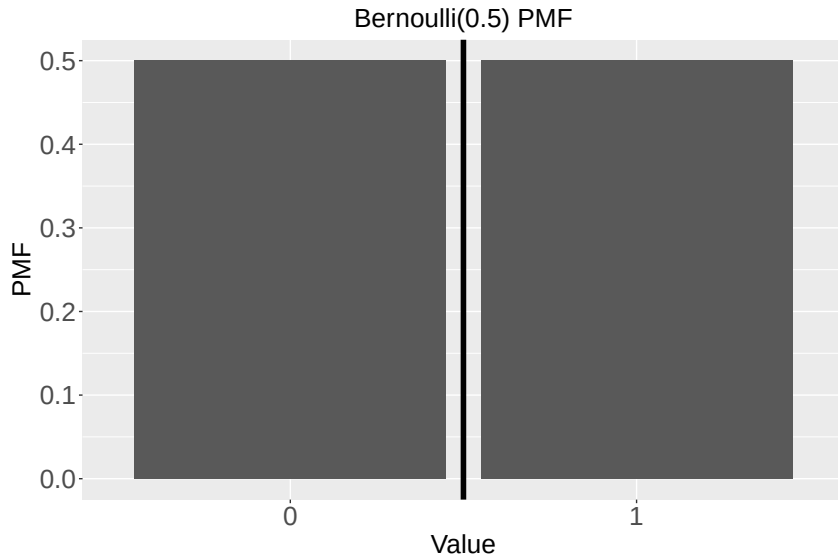
- ▶ A statistical analysis typically proceeds by selecting a PMF that seems to match the distribution of a sample.
- ▶ We rarely know the PMF exactly, but we assume it is from a parametric family of distributions.
- ▶ For example, $\text{Binomial}(10,0.5)$ and $\text{Binomial}(4,0.1)$ are different but both from the binomial family.
- ▶ A family of distributions have the same equation for the PMF but differ by some unknown parameters θ .
- ▶ We must estimate these parameters.

Example: $X \sim \text{Bernoulli}(\theta)$

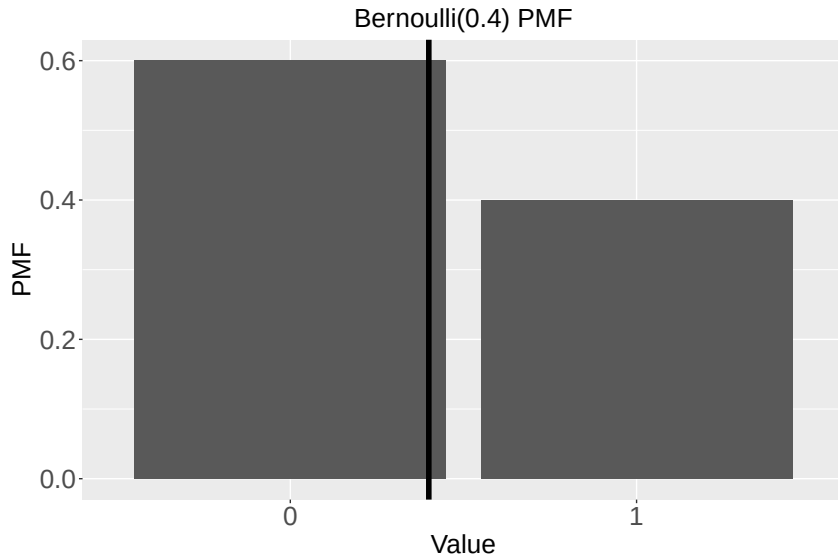
- ▶ Example: X is a success (1) or failure (0).
- ▶ Domain: $X \in \{0, 1\}$ (i.e., X is binary).
- ▶ PMF: $P(X = 0) = 1 - \theta$ and $P(X = 1) = \theta$.
- ▶ Parameter: $\theta \in [0, 1]$ is the success probability.
- ▶ Mean: $E(X) = \sum_x xf(x) = 0(1 - \theta) + 1\theta = \theta$.
- ▶ Variance:

$$V(X) = \sum_x (x - \theta)^2 f(x) = (0 - \theta)^2 (1 - \theta) + (1 - \theta)^2 \theta = \theta(1 - \theta).$$

Example: $X \sim \text{Bernoulli}(0.5)$



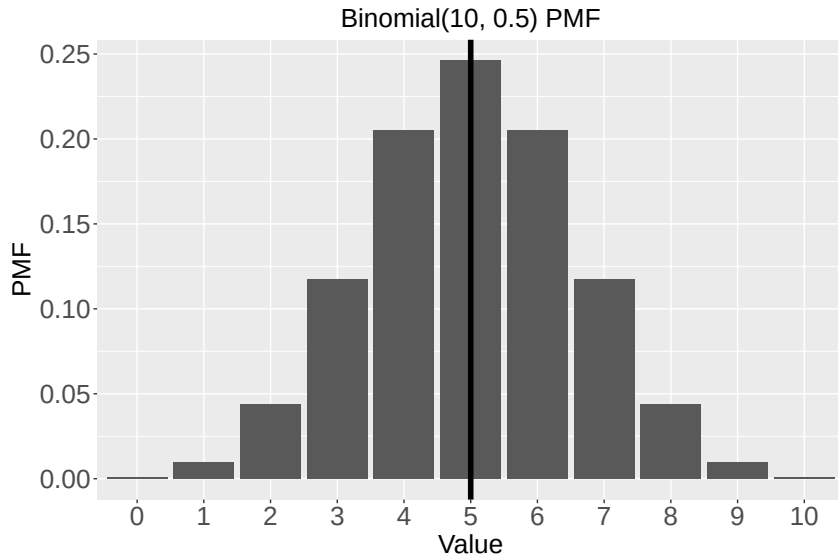
Example: $X \sim \text{Bernoulli}(0.4)$



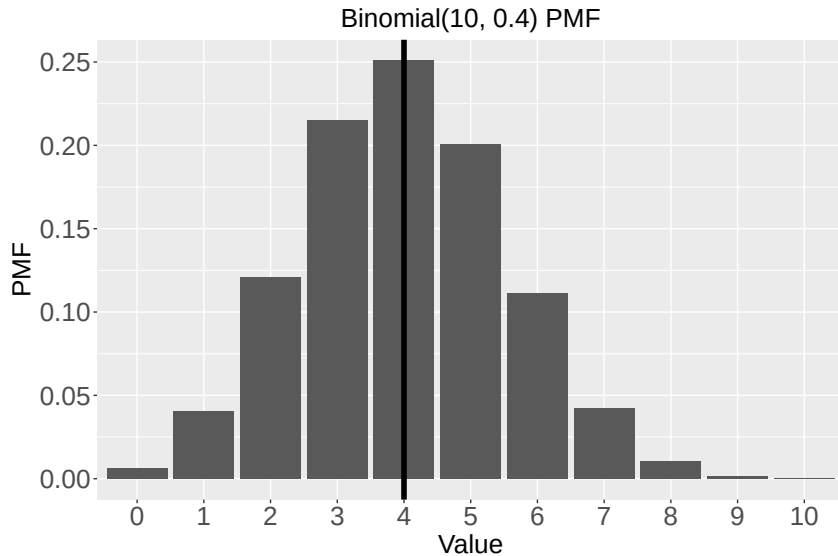
Example: $X \sim \text{Binomial}(N, \theta)$

- ▶ Example: X is a number of successes in N trials.
- ▶ Construction: $X = Y_1 + \dots + Y_N$ where $Y_1, \dots, Y_N \stackrel{iid}{\sim} \text{Bernoulli}(\theta)$.
- ▶ Domain: $X \in \{0, 1, \dots, N\}$.
- ▶ PMF: $P(X = x) = \binom{N}{x} \theta^x (1 - \theta)^{N-x}$.
- ▶ Parameter: $\theta \in [0, 1]$ is the success probability of each trial.
- ▶ Mean: $E(X) = \sum_{x=0}^N x f(x) = n\theta$.
- ▶ Variance: $V(X) = n\theta(1 - \theta)$.

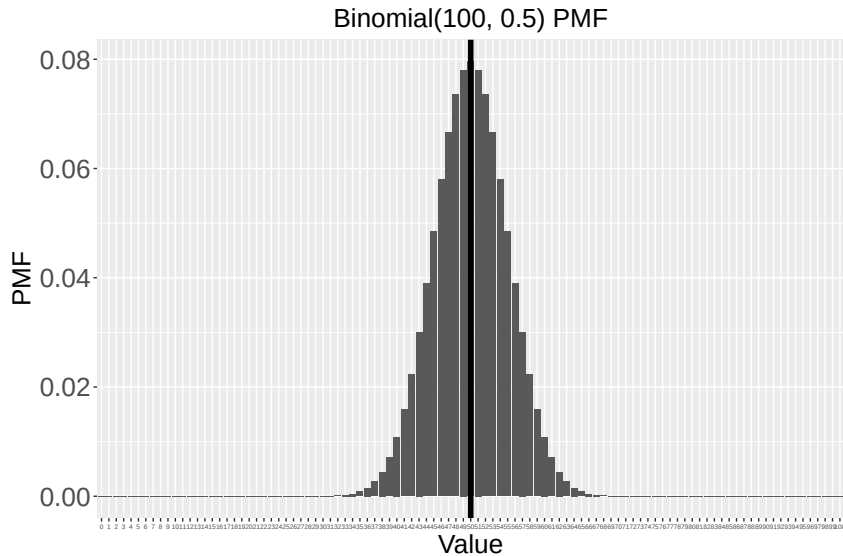
Example: $X \sim \text{Binomial}(10, 0.5)$



Example: $X \sim \text{Binomial}(10, 0.4)$



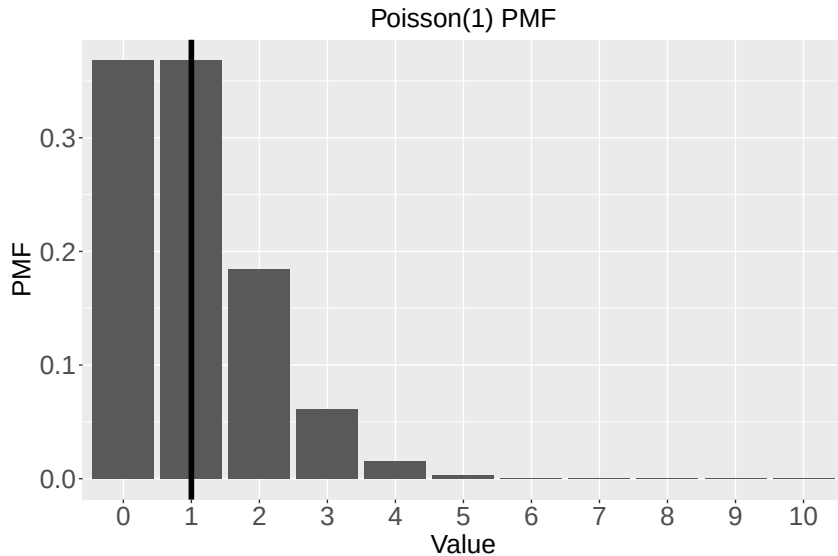
Example: $X \sim \text{Binomial}(100, 0.5)$



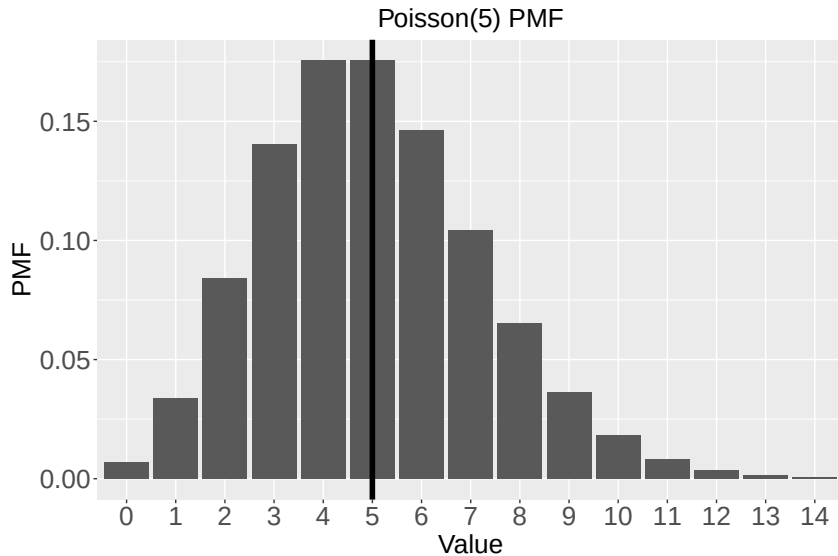
Example: $X \sim \text{Poisson}(N\theta)$

- ▶ Example: X is the number events that occur in N units of time
- ▶ Often the distribution is presented with $N = 1$.
- ▶ Domain: $X \in \{0, 1, 2, \dots\}$.
- ▶ PMF: $P(X = x) = \frac{\exp(-N\theta)(N\theta)^x}{x!}$.
- ▶ Parameter: θ is the expected number of events per unit of time
- ▶ Mean: $E(X) = N\theta$.
- ▶ Variance: $V(X) = N\theta$.

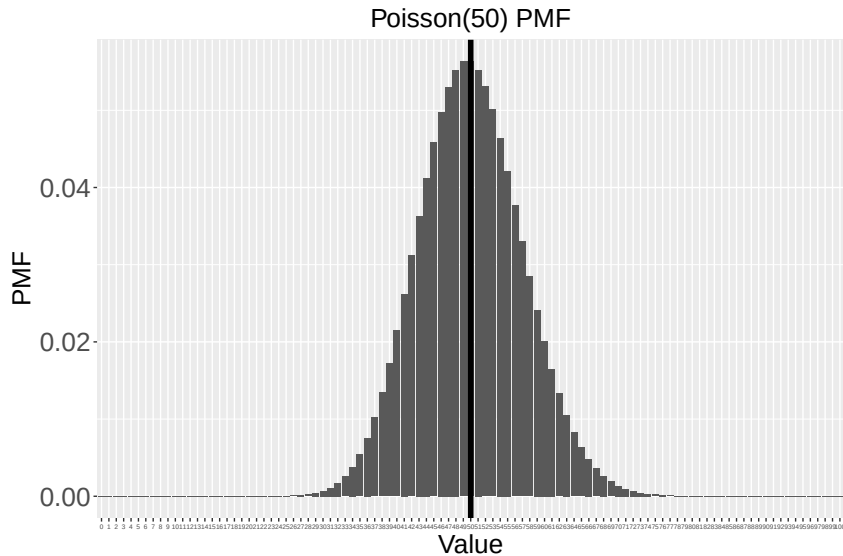
Example: $X \sim \text{Poisson}(1)$



Example: $X \sim \text{Poisson}(5)$



Example: $X \sim \text{Poisson}(50)$



Continuous univariate distributions

- ▶ If X is continuous we describe its distribution with the cumulative distribution function $F(x) \in [0, 1]$ or by probability density function (PDF) $f(x) \geq 0$.
- ▶ Here, $F(x) = \int_{-\infty}^x f(y)dy$ is a continuous function.
- ▶ Therefore, $P(X = x) = F(x) - \lim_{\delta \rightarrow 0} F(x - \delta) = 0$.
- ▶ Thus, the PMF is meaningless here.

Continuous univariate distributions

- ▶ Probabilities are computed as areas under the PDF curve

$$\text{Prob}(l < X < u) = \int_l^u f(x)dx.$$

- ▶ Therefore, $f(x)$ must satisfy $f(x) \geq 0$ and

$$\text{Prob}(-\infty < X < \infty) = \int_{-\infty}^{\infty} f(x)dx = 1.$$

Continuous univariate distributions

- ▶ The domain is the set of x values with $f(x) > 0$.
- ▶ The mean and the variance are defined similarly to the discrete case but with the sums replaced by integrals.

- ▶ The mean is

$$E(X) = \int xf(x)dx.$$

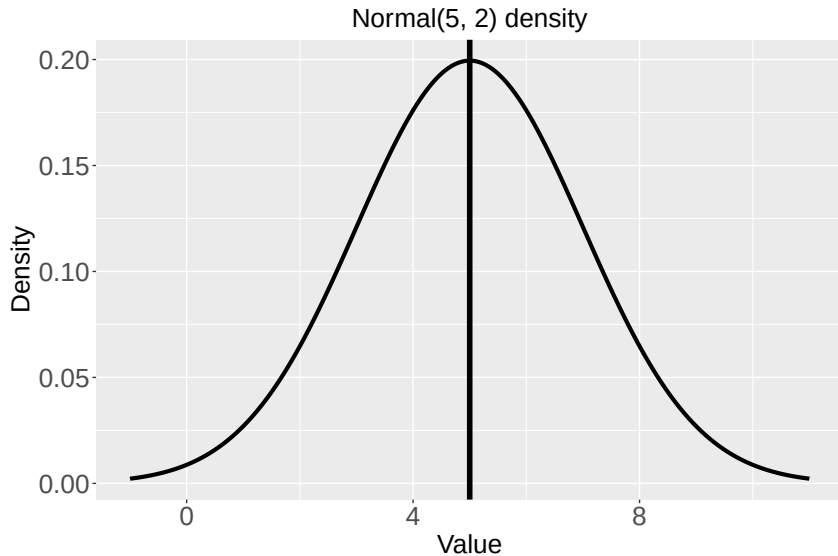
- ▶ The variance is

$$V(X) = \int [x - E(X)]^2 f(x)dx.$$

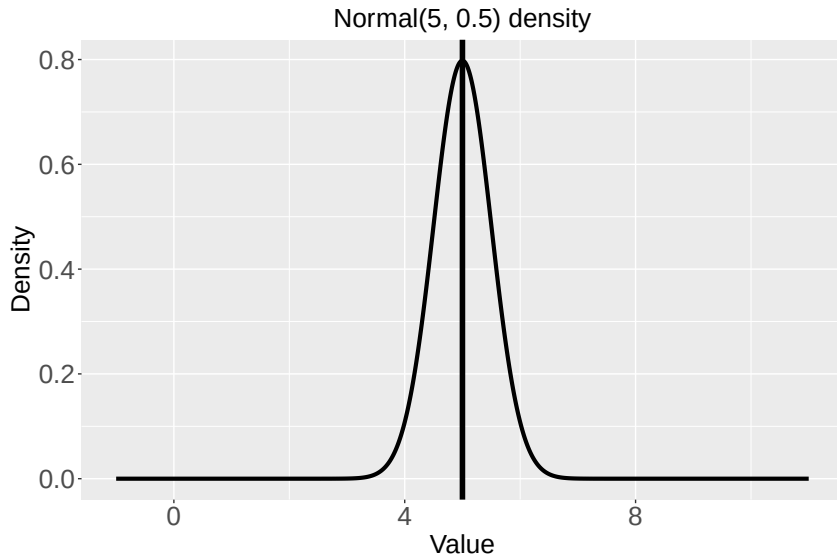
Example: $X \sim \text{Normal}(\mu, \sigma^2)$

- ▶ Example: X is the average temperature at a weather station on a certain day.
- ▶ Domain: $X \in (-\infty, \infty)$.
- ▶ PDF: $f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2} \left(\frac{x-\mu}{\sigma}\right)^2\right]$.
- ▶ Parameters: μ is the mean, $\sigma^2 > 0$ is the variance.
- ▶ Mean: $E(X) = \mu$.
- ▶ Variance: $V(X) = \sigma^2$.

Example: $X \sim \text{Normal}(5, 2^2)$



Example: $X \sim \text{Normal}(5, 0.5^2)$



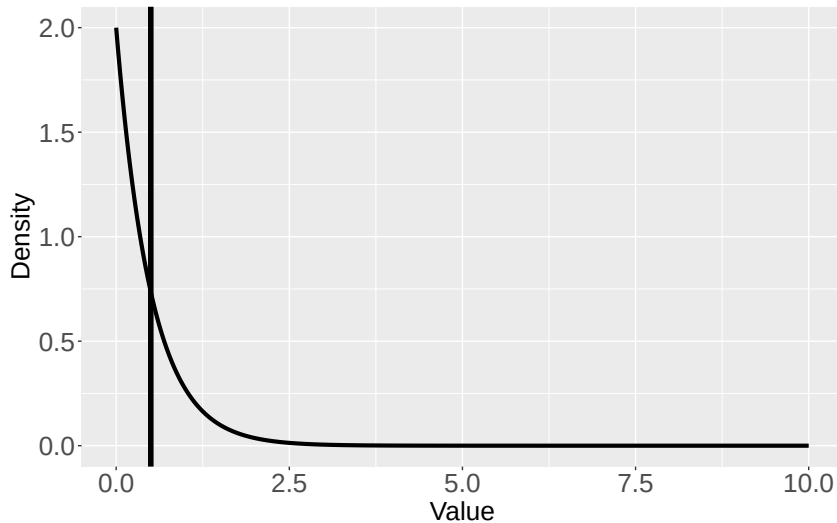
Example: $X \sim \text{Gamma}(a, b)$

- ▶ Example: X is a height of a baby.
- ▶ Domain: $X \in (0, \infty)$.
- ▶ PDF: $f(x) = \frac{b^a}{\Gamma(a)} x^{a-1} \exp(-bx)$.
- ▶ Parameters: $a > 0$ is the shape, $b > 0$ is the rate.
- ▶ Mean: $E(X) = \frac{a}{b}$.
- ▶ Variance: $V(X) = \frac{a}{b^2}$.
- ▶ Be careful: Sometimes the PDF is given as

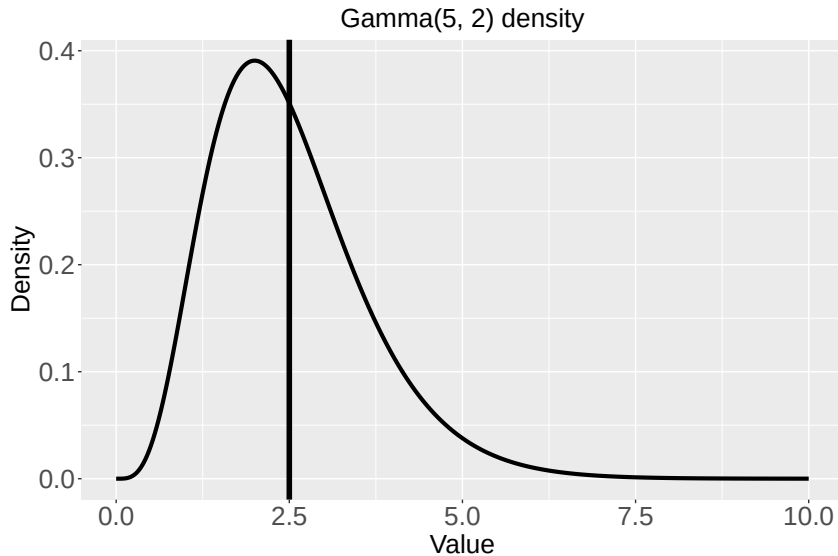
$$f(x) = \frac{1}{\Gamma(a)b^a} x^{a-1} \exp(-x/b).$$

Example: $X \sim \text{Gamma}(1, 2)$

Gamma(1, 2)=Exponential(2) density



Example: $X \sim \text{Gamma}(5, 2)$

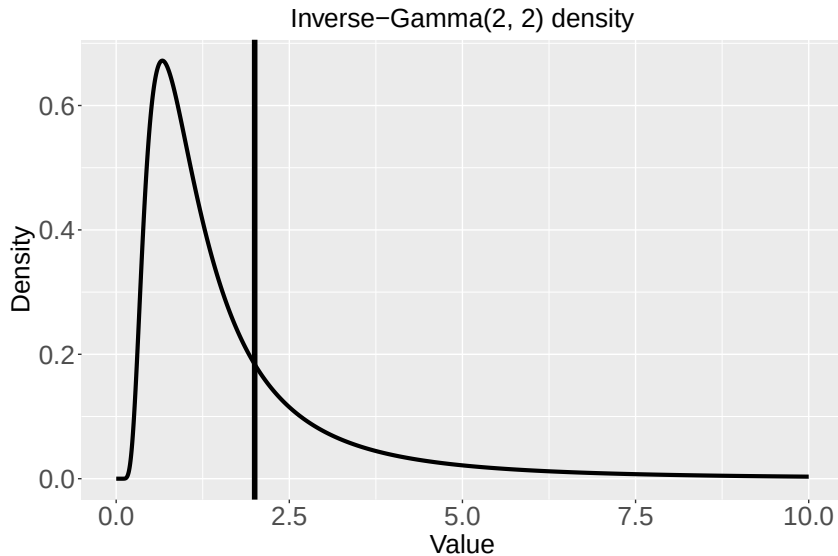


Example: $X \sim \text{Inverse-Gamma}(a, b)$

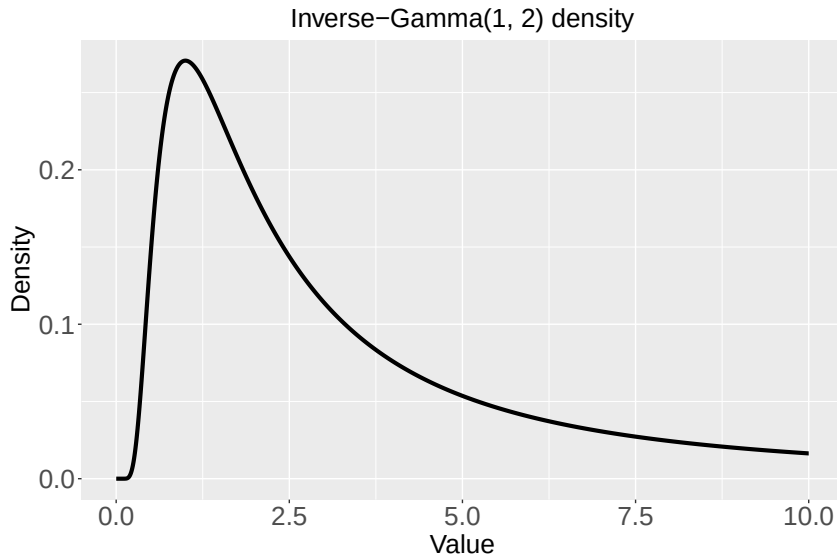
- ▶ If $Y \sim \text{Gamma}(a, b)$ and $X = 1/Y$, then $X \sim \text{Inverse-Gamma}(a, b)$.
- ▶ Domain: $X \in (0, \infty)$.
- ▶ PDF: $f(x) = \frac{b^a}{\Gamma(a)} x^{-a-1} \exp(-b/x)$.
- ▶ Parameters: $a > 0$ is the shape, $b > 0$ is the rate.
- ▶ Mean: $E(X) = \frac{b}{a-1}$ if $a > 1$.
- ▶ Variance: $V(X) = \frac{b^2}{(a-1)^2(a-2)}$ if $a > 2$.
- ▶ Be careful: Sometimes the PDF is given as

$$f(x) = \frac{1}{\Gamma(a)b^a} x^{-a-1} \exp[-1/(bx)].$$

Example: $X \sim \text{Inverse-Gamma}(2, 2)$



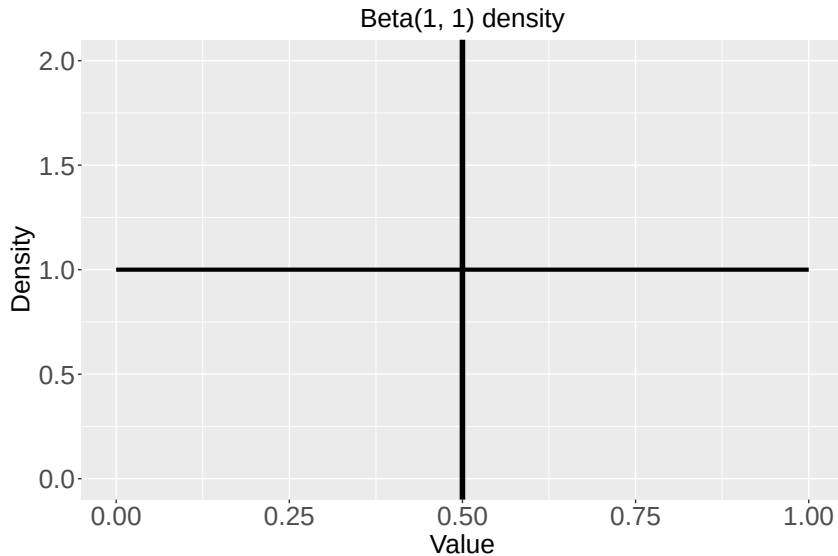
Example: $X \sim \text{Inverse-Gamma}(1, 2)$



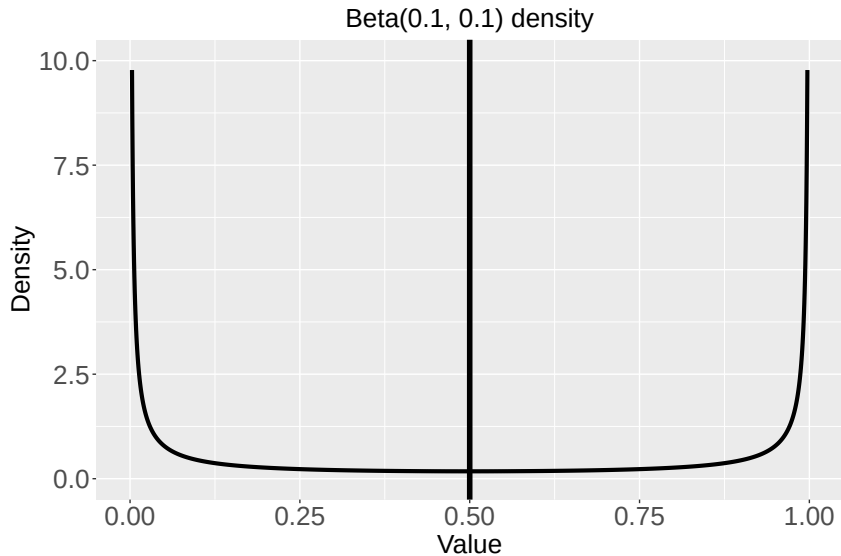
Example: $X \sim \text{Beta}(a, b)$

- ▶ Example: X is a probability.
- ▶ Domain: $X \in [0, 1]$.
- ▶ PDF: $f(x) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1} (1-x)^{b-1}$.
- ▶ Parameters: $a > 0$ and $b > 0$.
- ▶ Mean: $E(X) = \frac{a}{a+b}$.
- ▶ Variance: $V(X) = \frac{ab}{(a+b)^2(a+b+1)}$.

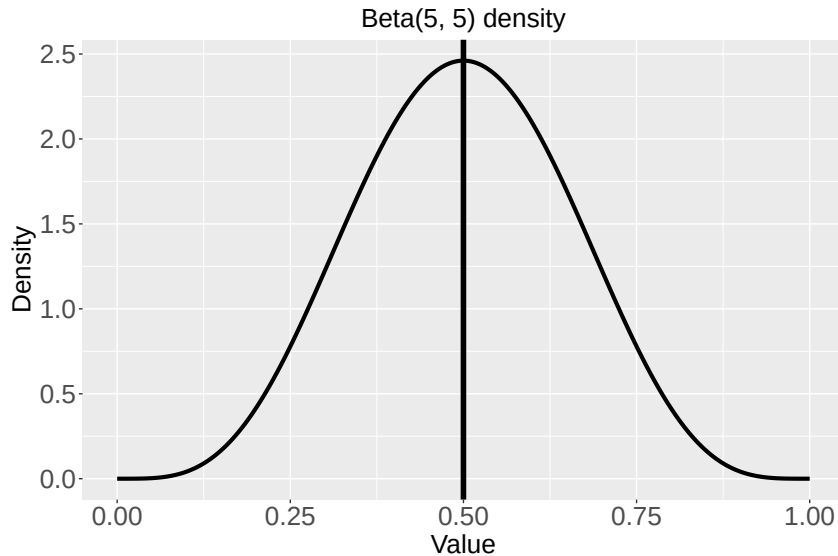
Example: $X \sim \text{Beta}(1, 1)$



Example: $X \sim \text{Beta}(0.1, 0.1)$



Example: $X \sim \text{Beta}(5, 5)$



Thank you!